

Study Notes: the Epicycle and the Vortical-Shwave Density

Derivations behind the Phase-0 PI-question analyses

(companion to `pi_questions_analysis.ipynb`)

2026-08-19

These notes collect, in print form, the two derivations written into the PI-questions notebook: (§1) the epicycle equations of motion and the reference solution used against the `epi_unif/epi_ring/epi_bndry` runs, and (§2) the envelope physics of the vortical-shwave density — why it peaks at the swing and decays afterwards — verified parameter-free against a fine-cadence rerun. §2.2 adds a step-by-step derivation of the linearized shearing-wave system itself, including how the background advection $-q\Omega x \partial_y$ becomes the time-dependent radial wavenumber $k_x(t)$. Conventions follow the code: $q \equiv -d \ln \Omega / d \ln R$ ($= 3/2$ Keplerian), background azimuthal flow $v_y = -q\Omega x$.

1 The epicycle: equations of motion and the reference solution

1.1 The shearing-sheet frame and its two forces

Work in a small Cartesian patch co-rotating with the disk at radius R_0 with angular frequency $\Omega \equiv \Omega(R_0)$: $x = R - R_0$ (radial), $y = R_0(\phi - \Omega t)$ (azimuthal), z vertical. Two inertial/gravity terms survive at first order in x/R_0 .

Tidal term. The effective potential in the rotating frame is $\Phi_{\text{eff}}(R) = \Phi(R) - \frac{1}{2}\Omega^2 R^2$. Circular-orbit balance gives $d\Phi/dR = R\Omega^2(R)$, and the local power law $\Omega \propto R^{-q}$ gives $\Omega^2(R_0+x) \simeq \Omega^2(1 - 2qx/R_0)$. Expanding to first order in x ,

$$\left. \frac{d\Phi_{\text{eff}}}{dR} \right|_{R_0+x} = (R_0+x)\Omega^2(R_0+x) - \Omega^2(R_0+x) \Big|_{\text{centrifugal}} = -2q\Omega^2 x + O(x^2/R_0), \quad (1)$$

i.e. a radial force $+2q\Omega^2 x \hat{\mathbf{x}}$: tidal stretching, outward beyond R_0 and inward inside. *This expansion is the linearization that defines the sheet; everything after it is exact within the sheet.*

Coriolis term. With $\boldsymbol{\Omega} = \Omega \hat{\mathbf{z}}$, $-2\boldsymbol{\Omega} \times \mathbf{v} = 2\Omega v_y \hat{\mathbf{x}} - 2\Omega v_x \hat{\mathbf{y}}$.

The momentum equations of the (unstratified) sheet are therefore

$$\frac{Dv_x}{Dt} = 2\Omega v_y + 2q\Omega^2 x - \frac{1}{\rho} \partial_x p, \quad \frac{Dv_y}{Dt} = -2\Omega v_x - \frac{1}{\rho} \partial_y p. \quad (2)$$

1.2 Equilibrium

$v_x = 0$, $v_y = -q\Omega x$ with uniform ρ, p balances exactly: $2\Omega(-q\Omega x) + 2q\Omega^2 x = 0$. This is the background that orbital advection (FARGO) splits off and advects analytically; the evolved azimuthal variable is the fluctuation $v'_y \equiv v_y + q\Omega x$.

1.3 Spatially uniform perturbation: an exact reduction

Take $v_x = v_x(t)$ and $v'_y = v'_y(t)$ uniform in space (the `ipert=1` initial condition: $v_x(0) = A$, $v'_y(0) = 0$), with ρ and p uniform. Then

- pressure gradients vanish;
- the fluctuation self-advection $\mathbf{v}' \cdot \nabla \mathbf{v}'$ vanishes *identically* — no linearization in the amplitude A is needed; the only surviving advection is $v_x \partial_x$ acting on the background $-q\Omega x$;
- continuity gives $\partial_t \rho = -\nabla \cdot (\rho \mathbf{v}) = 0$: the density stays *exactly* ρ_0 (the property the Q0.1 interface test exploits).

Substituting $v_y = -q\Omega x + v'_y$ into Eq. (2):

$$\dot{v}_x = 2\Omega(-q\Omega x + v'_y) + 2q\Omega^2 x = 2\Omega v'_y, \quad (3)$$

$$\dot{v}'_y - q\Omega v_x = -2\Omega v_x \implies \dot{v}'_y = -(2 - q)\Omega v_x, \quad (4)$$

where the $-q\Omega v_x$ on the left is the advection of the background shear.

1.4 The epicyclic oscillator and the reference solution

Combining the two equations,

$$\ddot{v}_x = -2(2 - q)\Omega^2 v_x \equiv -\kappa^2 v_x, \quad \boxed{\kappa^2 = 2(2 - q)\Omega^2} \quad (5)$$

the local form of the general epicyclic frequency $\kappa^2 = R^{-3} d(R^4 \Omega^2)/dR = (4 - 2q)\Omega^2$. For $q = 3/2$: $\kappa = \Omega$. With the code's initial condition,

$$v_x(t) = A \cos \kappa t, \quad v'_y(t) = \frac{\dot{v}_x}{2\Omega} = -\frac{A\kappa}{2\Omega} \sin \kappa t. \quad (6)$$

The orbit in (v_x, v'_y) space is a clockwise ellipse with axis ratio $\kappa/2\Omega$ ($= 1/2$ Keplerian); the displacement $x(t) = x_0 + (A/\kappa) \sin \kappa t$ is the classical retrograde epicycle.

1.5 The two diagnostics used against the runs

- **Direct trajectory:** volume-averaged $\langle \rho v_x \rangle / \langle \rho \rangle$ vs. $A \cos \kappa t$. Envelope differences here are dominated by accumulated RK2 *phase* drift.
- **Amplitude invariant:** eliminating t ,

$$\mathcal{A}(t) \equiv \sqrt{v_x^2 + \left(\frac{2\Omega}{\kappa}\right)^2 v_y'^2} = A = \text{const}, \quad (7)$$

conserved by the exact dynamics at *any* amplitude (it is the oscillator energy in disguise), so its drift isolates numerical damping/growth from phase error.

- For short runs the volume-integrated x -kinetic energy $\int \frac{1}{2} \rho v_x^2 dV = \frac{1}{2} \rho_0 A^2 \cos^2(\kappa t) V_{\text{box}}$ is used instead.

1.6 100-orbit results (2026-08-19 reruns)

run	v_x range/ A	invariant drift	$\max v_x - A \cos \kappa t /A$
uniform 128^2	$[-1.00059, +1.00045]$	+0.321%	0.124
ring SMR	$[-1.00001, +1.00000]$	+0.040%	0.031
bndry SMR	$[-1.00001, +1.00000]$	+0.040%	0.031

The two SMR runs are bit-identical to each other over all 100 orbits; the uniformity sweep again gives $\max |\rho - 1| = 0.0$ exactly in every float32 snapshot. Both error channels are pure RK2 time integration and scale like it: the SMR runs integrate the whole domain at half the timestep (global Δt set by the fine level), and the measured uniform/SMR ratios are $0.124/0.031 = 4.0 = (\Delta t_u/\Delta t_s)^2$ for phase error and $0.321/0.040 = 8.0 = (\Delta t_u/\Delta t_s)^3$ for the invariant drift — RK2’s per-unit-time phase error is $O(\Delta t^2)$ and its secular amplitude growth $O(\Delta t^3)$, both derived in §1.7 below.

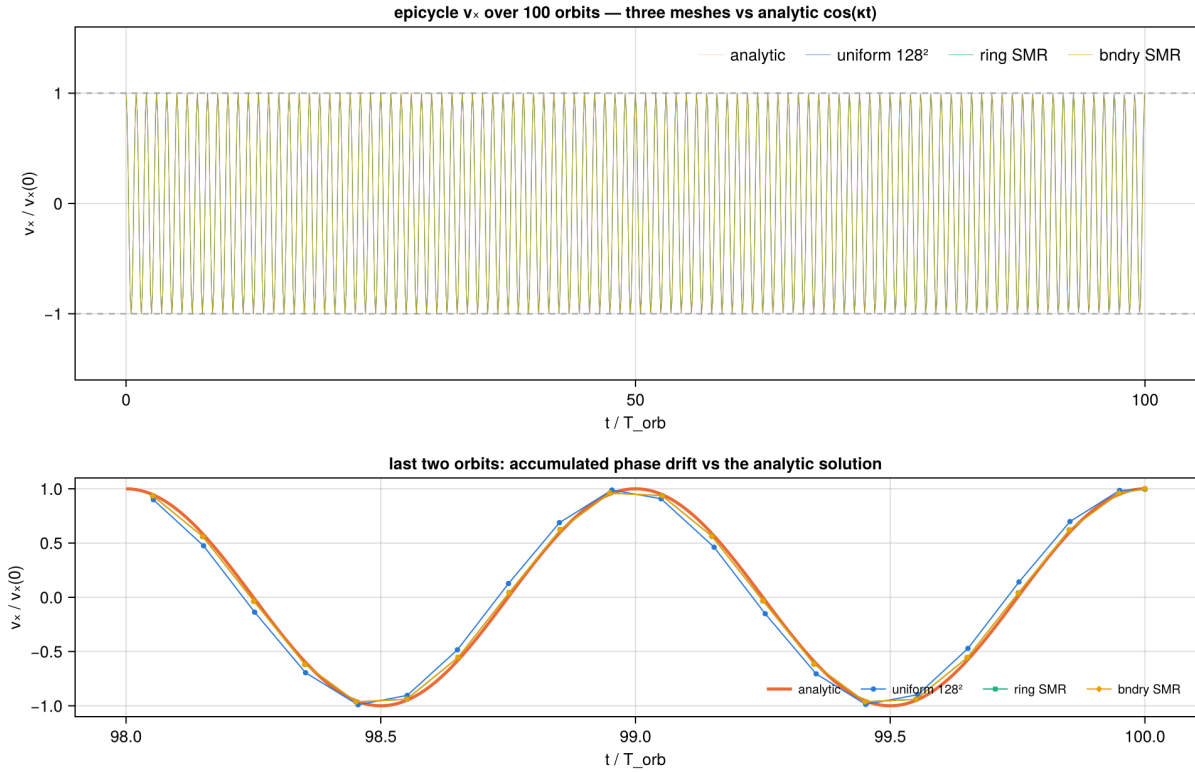


Figure 1: $v_x/v_x(0)$ over 100 orbits for the three meshes vs. the analytic $\cos \kappa t$ (top: full span; bottom: final two orbits). The uniform run leads in phase by $\sim 1/8$ period; the two SMR runs overlap (bit-identical) and hug the analytic curve.

1.7 Why the drifts: RK2 on a pure oscillator

Reduction to the model problem. Define the complex variable $u = v_x + i(2\Omega/\kappa)v'_y$. The epicycle pair (Eqs. 3–4) combines into exactly

$$\dot{u} = -i\kappa u, \quad (8)$$

whose exact solution rotates u on a circle at rate κ with $|u|$ constant — and $|u|$ is the amplitude invariant $\mathcal{A}$ of Eq. (7). “Invariant drift” is therefore precisely “what the integrator does to $|u|$ per step”. Because the uniform epicycle collapses the full PDE to this single ODE, the textbook stability analysis below applies to the simulation *literally*, with no spatial-discretization contamination.

One RK2 step. For any linear problem $\dot{u} = \lambda u$, one step of any two-stage second-order Runge–Kutta method (Heun, midpoint — AthenaK’s `rk2` included) gives exactly

$$u^{n+1} = R(z) u^n, \quad R(z) = 1 + z + \frac{z^2}{2}, \quad z = \lambda \Delta t, \quad (9)$$

the Taylor series of the true propagator e^z truncated after the z^2 term. The truncation is the whole story.

Amplitude: why growth, and why $(\omega \Delta t)^4/8$. Here $z = -i\theta$ with $\theta = \kappa \Delta t$, and the exact propagator $e^{-i\theta}$ has modulus exactly 1. The truncated polynomial does not:

$$|R|^2 = \left(1 - \frac{\theta^2}{2}\right)^2 + \theta^2 = 1 - \theta^2 + \frac{\theta^4}{4} + \theta^2 = 1 + \frac{\theta^4}{4} \implies |R| \simeq 1 + \frac{\theta^4}{8}. \quad (10)$$

The θ^2 terms cancel exactly (that is what second order buys: forward Euler has $|R|^2 = 1 + \theta^2$), leaving a *positive* remainder: each step multiplies the amplitude by $1 + (\kappa \Delta t)^4/8 > 1$. Geometrically, the exact step moves u along the circle; the RK2 step moves it along the parabola that matches the circle to second order, and a parabola hugging a circle from outside lands slightly *outside* it — a tiny energy injection, every step, always outward. RK2 is weakly *anti-dissipative* on oscillations: not noise that averages out, but a systematic bias. (RK4’s quartic lands slightly *inside*, $|R_4|^2 = 1 - \theta^6/72 + \dots$, weak damping; leapfrog/symplectic schemes land exactly on the circle and err only in phase.)

Accumulation and the Δt^3 law. Per step the relative gain $(\kappa \Delta t)^4/8$ is invisibly small (1.7×10^{-7} at $\Delta t = 0.0342$), but a run of length T takes $N = T/\Delta t$ steps and the gains compound with the same sign:

$$\frac{\Delta \mathcal{A}}{\mathcal{A}} = N \frac{(\kappa \Delta t)^4}{8} = \boxed{\frac{T \kappa^4 \Delta t^3}{8}} \quad (11)$$

— *linear in T* (secular; measured: +0.064% at 20 orbits $\rightarrow$ +0.321% at 100) and $\propto \Delta t^3$ per unit time (one power of Δt is spent on needing more steps). Halving the step cuts the drift by $2^3 = 8$: the SMR runs take exactly half the uniform step, and $0.321/0.040 = 8.0$.

Phase: the same logic, one order earlier. The argument of R is also slightly wrong: $\arg R = -(\theta + \theta^3/6 + \dots)$, i.e. the numerical rotation *overshoots* the true angle by $\theta^3/6$ per step — which is why the runs *lead* the analytic cosine in Fig. 1. Accumulated: $\Delta \varphi = N \theta^3/6 = T \kappa^3 \Delta t^2/6 \propto \Delta t^2$, hence the factor 4 at half the step ($0.124/0.031 = 4.0$).

The formulas predict the absolute numbers. With $T = 200\pi$, $\kappa = 1$, and the measured timesteps ($\Delta t = 0.0342$ uniform, 0.0171 SMR) — no fitting anywhere:

quantity	predicted	measured
uniform invariant drift, $T\kappa^4\Delta t^3/8$	+0.315%	+0.321%
SMR invariant drift	+0.039%	+0.040%
uniform phase offset, $T\kappa^3\Delta t^2/6$	0.123 rad	0.124
SMR phase offset	0.0307	0.0308

All four agree to $\sim 2\%$. This is the Q0.1 conclusion from yet another angle: the entire error budget of these runs is the time integrator behaving exactly like its textbook analysis — the mesh, its interfaces, and the shear machinery contribute nothing measurable.

2 The vortical-shwave density: peaks, nulls, and decay

Context (Q0.2): the density panels of the shwave comparison movie flicker frame-to-frame. The flicker is temporal aliasing of the free acoustic ringing; this section derives the smooth envelope underneath and what sets it. The derivation is self-contained, but the framework is the “shwave” linear theory of Johnson & Gammie (2005a), and the test problem is the code-verification setup of Johnson & Gammie (2005b, their Fig. 1) — both in `docs/`.

2.1 Single shearing wave

Linear isothermal perturbations on the shear, as one shearing wave $\propto \exp[i(k_x(t)x + k_y y)]$ with

$$k_x(t) = k_{x0} + q\Omega k_y t, \quad (12)$$

so the shear tilts the wave through the *swing* $k_x = 0$ at $t_{\text{swing}} = -k_{x0}/(q\Omega k_y)$. (Movie configuration: 64^2 , box 0.5^2 , $c_s = 1$, $(n_{wx}, n_{wy}) = (-8, 2)$, $\text{amp} = 10^{-4}$, so $k_{x0} = -32\pi$, $k_y = 8\pi$, k_x/k_y runs $-4 \rightarrow 0 \rightarrow +3.5$ and $t_{\text{swing}} = 8/3$.) The Fourier amplitudes obey

$$\dot{v}_x = 2\Omega \hat{v}_y - ik_x c_s^2 \hat{\delta}, \quad \dot{v}_y = -(2-q)\Omega \hat{v}_x - ik_y c_s^2 \hat{\delta}, \quad \dot{\hat{\delta}} = -i(k_x \hat{v}_x + k_y \hat{v}_y), \quad (13)$$

with $\hat{\delta} \equiv \hat{\rho}/\rho_0$ and $k^2(t) = k_x^2(t) + k_y^2$. The next subsection derives Eqs. (12)–(13) from the sheet equations; a reader who takes them as given can skip to §2.3.

2.2 Where the shearing-wave system comes from

Derivation of Eqs. (12)–(13) from the sheet equations. The only step here that is not routine is how the background advection $-q\Omega x \partial_y$ — whose explicit factor of x spoils an ordinary plane wave — is absorbed into a *time-dependent* radial wavenumber.

Linearization, term by term. Add continuity to the sheet momentum equations (2) with $p = c_s^2 \rho$ (isothermal), and split $\rho = \rho_0(1 + \delta)$, $\mathbf{v} = \mathbf{v}^0 + \mathbf{v}'$ about the equilibrium $\mathbf{v}^0 = -q\Omega x \hat{\mathbf{y}}$ of §1.2, keeping first order in $(\delta, \mathbf{v}')$. The advective term contributes three pieces,

$$(\mathbf{v} \cdot \nabla) \mathbf{v} = \underbrace{(\mathbf{v}^0 \cdot \nabla) \mathbf{v}^0}_{=0} + \underbrace{(\mathbf{v}^0 \cdot \nabla) \mathbf{v}'}_{-q\Omega x \partial_y \mathbf{v}'} + \underbrace{(\mathbf{v}' \cdot \nabla) \mathbf{v}^0}_{-q\Omega v'_x \hat{\mathbf{y}}} + O(v'^2), \quad (14)$$

of which the second is advection *of* the perturbation *by* the background (the term of interest below) and the third is advection of the background by the perturbation — it hits only the y -equation, because $\mathbf{v}^0$ has only a y -component, and it is the same term that turned -2Ω into $-(2-q)\Omega$ in Eq. (4). In the x -equation the Coriolis and tidal backgrounds cancel, $2\Omega(-q\Omega x + v'_y) + 2q\Omega^2 x = 2\Omega v'_y$ — that cancellation *is* the equilibrium condition — and the pressure gradient linearizes to $-c_s^2 \nabla \delta$ (uniform background pressure, so $1/\rho$ may be evaluated at ρ_0). With $\nabla \cdot \mathbf{v}^0 = 0$, continuity gives $\partial_t \delta - q\Omega x \partial_y \delta = -\nabla \cdot \mathbf{v}'$. All three equations therefore carry the *same* linearized advective operator $D_t \equiv \partial_t - q\Omega x \partial_y$:

$$D_t v'_x = 2\Omega v'_y - c_s^2 \partial_x \delta, \quad D_t v'_y = -(2-q)\Omega v'_x - c_s^2 \partial_y \delta, \quad D_t \delta = -(\partial_x v'_x + \partial_y v'_y). \quad (15)$$

Why a fixed- k Fourier mode fails. For $f \propto e^{i(k_x x + k_y y)}$ with constant k_x , the offending term is $-q\Omega x \partial_y f = -i q\Omega k_y x f$: the explicit x maps the mode onto $x \times (\text{mode})$, a *different* function, so plane waves are not eigenfunctions of D_t . Since $x \leftrightarrow i \partial / \partial k_x$ under Fourier transform, this term is an advection in k -space, $\partial_t \hat{f} + q\Omega k_y \partial \hat{f} / \partial k_x = (\text{local terms})$, whose characteristics are $dk_x/dt = q\Omega k_y$. Modes drift along the k_x axis at a fixed rate; the shearing-wave ansatz simply rides that characteristic.

The shearing-wave ansatz (the load-bearing step). Put the time dependence in the wavenumber itself, $f(x, y, t) = \hat{f}(t) e^{i\varphi}$ with $\varphi = k_x(t)x + k_y y$. Then ∂_t at *fixed position* picks up two pieces — the changing amplitude and the rotating phase — and

$$D_t f = \left[\dot{\hat{f}} + i \dot{k}_x x \hat{f} \right] e^{i\varphi} - i q\Omega k_y x \hat{f} e^{i\varphi} = \left[\dot{\hat{f}} + i x (\dot{k}_x - q\Omega k_y) \hat{f} \right] e^{i\varphi}. \quad (16)$$

The x -dependent piece — exactly what broke the fixed- k mode — vanishes for *all* x if and only if $\dot{k}_x = q\Omega k_y$, i.e. Eq. (12), and then $D_t f = \dot{\hat{f}} e^{i\varphi}$: the entire background advection has been absorbed into the drifting wavenumber. The geometric content is that surfaces of constant phase are *material* surfaces of the background flow — along a background trajectory (x fixed, $y(t) = y_0 - q\Omega x t$),

$$\varphi(t) = (k_{x0} + q\Omega k_y t) x + k_y (y_0 - q\Omega x t) = k_{x0} x + k_y y_0 = \text{const}, \quad (17)$$

so “ d/dt at fixed phase” is the shear tilting the crests, and tilting crests *is* winding up k_x . Equivalently, in shearing coordinates $y' = y + q\Omega t x$ this is an ordinary Fourier mode $e^{i(k_{x0} x' + k_y y')}$, and expanding the exponent returns $k_x(t)x + k_y y$ — the same relation the FFT solver applies to the transform of the rolled (sheared-coordinate) density, $k_x^{\text{phys}} = k'_x + q\Omega t (L_x/L_y) k_y$, where k'_x, k_y are integer mode indices and the L_x/L_y is the conversion between them.

Collapse to Eq. (13). With the ansatz in force, $D_t \rightarrow d/dt$, $\partial_x \rightarrow i k_x(t)$, $\partial_y \rightarrow i k_y$, and the common factor $e^{i\varphi}$ divides out of Eq. (15) — leaving precisely Eq. (13), with $\hat{v}_x, \hat{v}_y$ the amplitudes of the fluctuations v'_x, v'_y . These ODEs are exact within linear theory: no WKB or slowly-varying assumption has been made (that enters only at Eq. (21)), and the sole approximation remains the dropping of $O(v'^2)$.

Two checks. (i) *Uniform limit:* at $k_{x0} = k_y = 0$ the pressure and compression terms vanish and Eq. (13) reduces to the epicycle pair of Eqs. (3)–(4) — §1 is the $k \rightarrow 0$ member of this family. (ii) *Potential vorticity:* differentiating ξ of Eq. (18) below and substituting Eq. (13), the c_s^2 terms cancel pairwise and the rotational terms collect to $i k_y \hat{v}_y [q - 2 + (2 - q)] \Omega = 0$ — but only because $\dot{k}_x = q\Omega k_y$, so PV conservation is itself a consistency check on the ansatz.

2.3 The carrier: conserved potential vorticity

The perturbation potential vorticity

$$\xi = i(k_x \hat{v}_y - k_y \hat{v}_x) - (2 - q) \Omega \hat{\delta} \quad (18)$$

is *exactly* conserved by Eqs. (13) (direct differentiation, using $\dot{k}_x = q\Omega k_y$). In the nearly incompressible regime the vortical wave is $\hat{v}_x = i k_y \hat{\psi}$, $\hat{v}_y = -i k_x \hat{\psi}$ with $\xi \simeq k^2 \hat{\psi}$, hence

$$|\mathbf{v}| = \frac{\xi}{k(t)} : \quad (19)$$

the velocity amplitude peaks where $k(t)$ is smallest — the swing — and decays as $1/k \propto 1/t$ afterwards as the shear rewinds the wave to ever finer radial scale. **Transient amplification, no instability:** the PV budget is fixed and merely gets spread over growing k . In component form, $\hat{v}_x = ik_y \hat{\psi} = ik_y \xi/k^2 \propto (1 + \tau^2)^{-1}$ with $\tau = k_x/k_y$ — exactly eq. (9) of Johnson & Gammie (2005b), the amplitude law the validation document verifies to $< 0.5\%$ of peak on all three meshes.

2.4 The density: a driven acoustic oscillator

Differentiating the continuity relation in Eq. (13) once and substituting the momentum equations,

$$\ddot{\delta} + c_s^2 k^2(t) \hat{\delta} = S(t), \quad S = -2i\Omega[(q-1)k_y \hat{v}_x + k_x \hat{v}_y] = \frac{2\Omega \xi [(q-1)k_y^2 - k_x^2(t)]}{k^2(t)}, \quad (20)$$

an acoustic oscillator at the instantaneous (WKB) frequency $c_s k(t)$, driven by the Coriolis/shear forces acting on the vortical flow. Its solution is a **slaved response plus free ringing**:

- *Slaved* (adiabatic; valid since $c_s k \gg \Omega$ throughout):

$$\hat{\delta}_{\text{siv}}(t) = \frac{S}{c_s^2 k^2} = \frac{2\Omega \xi [(q-1)k_y^2 - k_x^2(t)]}{c_s^2 k^4(t)}, \quad \xi = \text{amp} \cdot \frac{k_0^2}{k_y}, \quad (21)$$

with $k_0 = k(0)$ (the pgen's amp is the v_x amplitude).

- *Free ringing* at $\omega = c_s k(t)$, launched at $t = 0$ because the pgen initializes $\delta\rho = 0$ rather than $\hat{\delta}_{\text{siv}}(0)$ (and re-pumped near the swing, where the drive varies fastest); WKB amplitude $\propto k^{-1/2}$. This is what the movie's 0.1-cadence frames undersample (2.4–2.9 beats of $|\hat{\delta}|$ per frame at $t \approx 5$), producing the stroboscopic flicker.

2.5 Envelope anatomy — all parameter-free

Writing $u = k_x^2/k_y^2$, Eq. (21) $\propto |u - (q-1)|/(u+1)^2$:

- a **local maximum at** $k_x = -\sqrt{2} k_y$ ($u = 2$; here $t = 1.72$);
- **nulls at** $k_x^2 = (q-1)k_y^2$ (here $t = 2.20$ and 3.14): the tilt angles at which the two Coriolis/shear force projections in S cancel;
- the **absolute peak at the swing** ($k_x = 0$, $t = 8/3$), where $1/k^4$ is maximal: $\hat{\delta}_{\text{peak}} = 2\Omega\xi(q-1)/(c_s^2 k_y^2) = 0.68 \times 10^{-4}$ for the movie configuration (measured: 0.76×10^{-4} , the excess being ringing on top). The density peaks at the swing for the same reason the velocity does — minimum k — but more sharply (k^{-4} vs. k^{-1}): pressure resists compression least when the wave is least wound;
- **decay as** $t \rightarrow \infty$: the slaved part falls as $2\Omega\xi/(c_s k_x)^2 \propto t^{-2}$ (stiffening pressure vs. weakening vortical drive), the ringing as $k^{-1/2} \propto t^{-1/2}$, so the late band is ringing-dominated. On longer horizons the winding drives the radial wavelength to the grid scale and numerical diffusion ($\propto k^2$) extinguishes everything super-exponentially. Nothing can grow: without self-gravity there is no feedback channel (contrast the `swing_selfgrav` runs at $4\pi G > 2$, where the crest grows instead of bouncing).

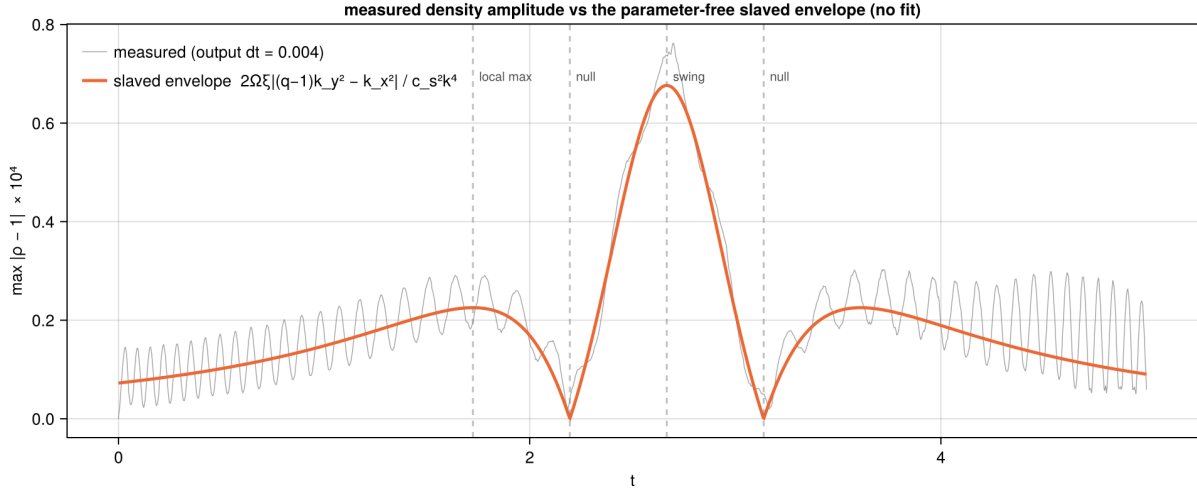


Figure 2: The measured density amplitude of the uniform vortical-shwave run (gray; output $dt = 0.004$) against the slaved envelope of Eq. (21) (orange) — *no fitted parameters*. Predicted local max ($t = 1.72$), nulls ($t = 2.20, 3.14$), swing peak ($t = 2.67, 0.68 \times 10^{-4}$) and the t^{-2} tail all line up; the fast oscillation around the envelope is the free acoustic ringing.

References

- Binney, J., & Tremaine, S. 2008, *Galactic Dynamics* (2nd ed.; Princeton), §3.2.3 — textbook epicycle theory behind §1.
- Goldreich, P., & Lynden-Bell, D. 1965, MNRAS, 130, 125 — shearing waves and swing amplification.
- Johnson, B. M., & Gammie, C. F. 2005a, ApJ, 626, 978 — linear “shwave” theory in the compressible shearing sheet; the linearized system of §2 (docs/Johnson_2005_ApJ_626_978.pdf; cited by the `shwave.cpp` problem generator).
- Johnson, B. M., & Gammie, C. F. 2005b, ApJ, 635, 149 — the vortical shwave as a code-verification test (their Fig. 1 and eq. 9) (docs/Johnson_2005_ApJ_635_149.pdf).
- Bodo, G., et al. 2005, A&A, 437, 9 — density-wave emission by vortical perturbations in Keplerian shear (the slaved-response/ringing physics of §2 in a broader setting).

The derivations themselves are worked directly from the linearized shearing-sheet equations (no formulas transcribed); the references anchor the framework, the test problem, and the cross-checks.

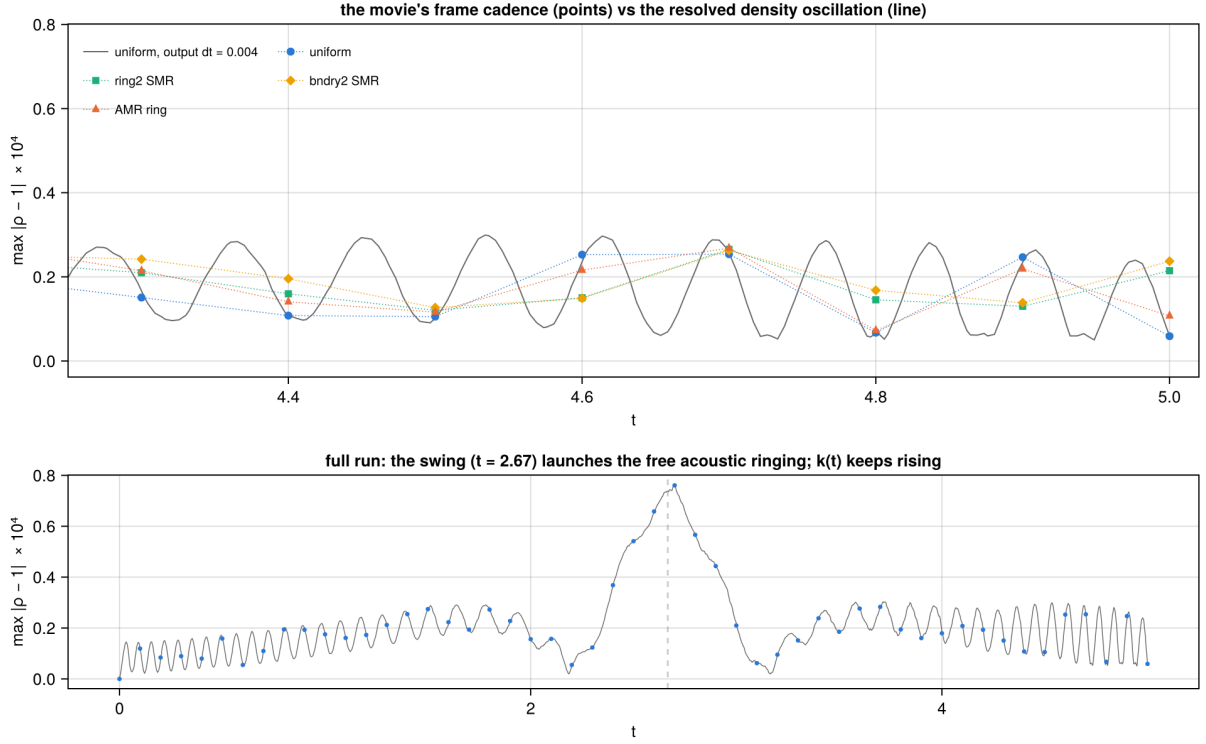


Figure 3: Why the movie flickers: the four runs' 0.1-cadence frame samples (points) hop across the resolved ringing (gray line) at quasi-random phases — 2.4–2.9 beats of $|\rho - 1|$ elapse between consecutive frames near $t = 5$. Bottom: full run; the ringing slows at the swing (k minimum) and accelerates as the wave rewinds.